Turn-Aware Streaming ASR in One Small Open Model: Semantic End-of-Turn and Context-Biased Transcription, with the Training Recipe

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Abstract

Every voice agent must answer one question in real time: *has the user finished talking?* The deployed default — a voice-activity detector plus a silence timeout — answers it from acoustics alone, and therefore cannot answer it well. A user dictating a phone number pauses between digit groups, and the timeout interrupts them mid-number; a user who says "hello, I'm still here" has finished the instant the words end, and the timeout keeps them waiting anyway. Both failures have one cause: end-of-turn is a property of *what was said*, not of how long the channel has been quiet. Frontier commercial systems have accordingly moved the decision into the recognizer — Deepgram's Flux, OpenAI's `semantic_vad` — but every member of this *turn-aware* class is closed, and the open landscape offers only pieces: Kyutai STT ships a semantic-VAD head with no training recipe, NVIDIA's Parakeet-EOU emits an end-of-utterance token with no published account of how it is trained, and open turn *detectors* (Smart Turn, LiveKit, TEN) are separate models that do not transcribe. We present, to our knowledge, the first open system that combines streaming transcription, an in-transcript semantic end-of-turn event, measured context biasing, and a fully published training recipe: Qwen3-ASR-0.6B, LoRA-fine-tuned on one GPU from $\sim$12k synthesized examples to emit turn markers inside its transcription stream, with a text context prefix biasing recognition throughout the session. On a deployment-matched streaming benchmark the model reaches 0.97 boundary recall at 0.39 s median latency with 0.3 false fires per speech-minute — a silence timeout needs 3$\times$ the latency and 5$\times$ the false fires to match that recall — and the context prefix lifts hotword recall by +28.9 pp on entity-dense Earnings-22 audio, an advantage that stays flat as the session grows; the fine-tune's offline WER cost is measured and attributed by ablation. Probes built from the two motivating scenarios show the recipe extends to dictation and spelling in *one* model: added schemas cut premature fires during phone-number dictation from 4.8 to 0.24 per number (better than any timeout) and ground spelled emails in a user profile, while distractor-ratio training — context and audio deliberately disagree, target follows the audio — drives wrong-profile intrusion from 40% to zero, all without a deployment-time checkpoint swap. The whole system runs on stock vLLM at 84 ms per 0.5 s chunk. The recipe's central lesson is transferable: obvious offline supervision demands *clairvoyance* — labels that depend on audio after the decision point — and it manufactures training oscillation and phantom accuracy trade-offs; one causal labeling rule, with nothing else changed, removes both. Weights, recipe, and benchmark are released.

Code: https://github.com/19PINE-AI/turn-aware-asr | Website: https://01.me/research/turn-aware-asr

1 Introduction

A voice agent that cannot tell when you have finished speaking is unusable in two symmetric ways: it interrupts you mid-thought, or it stares at you after you stop. As full-duplex assistants proliferate, this endpointing decision — fire an end-of-turn event now, or wait — has become the load-bearing interaction primitive, and the industry default remains a cascade: a voice-activity detector (VAD) plus a fixed silence timeout, feeding a separate recognizer. The cascade's problem is structural, not incremental. Humans exchange turns with median gaps around 200 ms [Stivers et al., 2009, Levinson and Torreira, 2015], while deployed silence timeouts sit at 500–1000 ms [Skantze, 2021]; yet pauses *inside* turns routinely exceed pauses *between* turns, so no threshold exists that is both fast and safe (§2). The information that resolves the ambiguity — is this thought complete? — is in the words, and the words live inside the recognizer.

Frontier commercial systems have drawn the conclusion and moved end-of-turn *into* the ASR model: Deepgram's Flux emits `EagerEndOfTurn` events from the recognizer's own forward pass [Deepgram, 2026], and OpenAI's Realtime API offers a `semantic_vad` mode that waits longer after a trailing "ummm" than after a completed sentence [OpenAI, 2025]. We call this class **turn-aware streaming ASR**: one model that transcribes while deciding, from semantics and silence jointly, whether the turn is over. Every commercial member of the class is closed — no weights, no recipe, no account of what makes the training work. The open landscape, surveyed in §12, offers pieces but not the combination: Kyutai STT ships an open-weights semantic-VAD head with no published training procedure [Kyutai Labs, 2025]; NVIDIA's Parakeet-EOU streams an end-of-utterance token with no published account of how the token's labels are derived [NVIDIA, 2026]; Smart Turn opens a full recipe but is a standalone endpoint classifier that produces no transcript [Pipecat team, 2025]; LiveKit's and TEN's detectors classify text downstream of a separate recognizer [LiveKit, 2025, TEN Team, 2025].

This paper opens the class. We LoRA-fine-tune a small open unified ASR model (Qwen3-ASR-0.6B, Shi et al., 2026) to emit end-of-turn marker tokens inside its transcription stream, so that one model transcribes, detects end-of-turn, and accepts a text context prefix — user profile, hotword list — that biases recognition for the whole session. Training uses ∼12k examples synthesized from public corpora and takes hours on one GPU; serving runs on stock vLLM at 84 ms per 0.5 s chunk. To our knowledge this is the first open system combining streaming transcription, an in-transcript semantic end-of-turn event, measured context biasing, and a fully published training recipe — data construction, label rules, and code.

The headline behavior is Figure 1. On a deployment-matched benchmark that replays real conversational timelines (§6), the model reaches 0.97 boundary recall at 0.39 s median latency with 0.3 false fires per speech-minute — and because it keys on *semantic completeness*, it escapes the one-dimensional latency-vs-false-fire curve that every silence timeout is confined to. A fresh, 2×-larger test set drawn after all development ended confirms the operating point (0.92 recall, 0.42 s, 0.2 false/min). The same model's context prefix lifts hotword recall by +28.9 pp

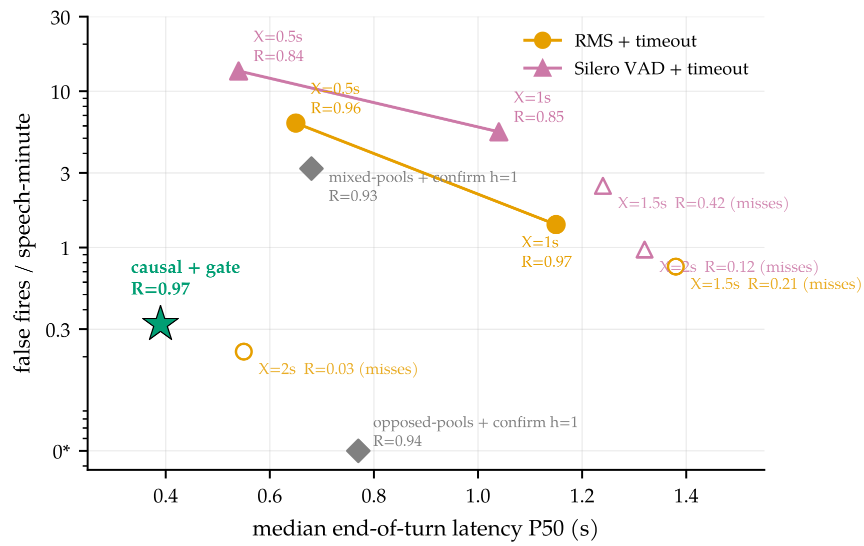

Figure 1. Semantic endpointing escapes the timeout curve. Median end-of-turn latency vs false fires (log scale; 0* = zero plotted at 0.05) on identical audio, detector, and scoring (§6). A silence timeout X can only slide along its curve: matching our model's recall takes X=1.0 s — 3× the latency, ∼5× the false fires; the fastest usable setting (X=0.5 s) produces 21× the false fires; $X \geq 1.5$ s leaves the recall tolerance window entirely (open markers) — the assistant that waits too long. The Silero cascade [Silero Team, 2021] is dominated by plain RMS on this close-talk channel. The causally-supervised model (§4) sits strictly inside the whole family because it reads *completeness*: it fires 0.39 s after a finished thought and holds through a 1.4 s mid-sentence pause.

on entity-dense earnings-call audio — the first quantitative context-biasing numbers on this model family, whose technical report describes the capability but publishes none [Shi et al., 2026] — and the advantage stays flat as the session grows (§8).

Getting the endpointing to work took nine supervision recipes, and the failure mode that consumed the first eight is, we believe, the reason no training account of this class has been published: it is genuinely easy to get wrong, in a way that masquerades as a fundamental capability trade-off. Offline corpora are cut at exactly the boundaries a streaming model must detect, and offline transcripts record who spoke next; supervision built naively on either demands *clairvoyance* — labels whose correct value depends on audio after the decision point. The result was training runs that oscillated between a model that always fired and one that never did, and an apparent recall-vs-precision Pareto frontier across checkpoints that dissolved the moment we rebuilt the labels under one causal rule, changing nothing else (§5). We distill the lesson into a definition, two mechanisms, and a four-question checklist (§11) that applies to any streaming decision trained from offline logs.

Our contributions:

1. **An open turn-aware streaming ASR system** (§3): one small model that transcribes, emits semantic end-of-turn events in-stream, and takes a biasing context prefix; weights, training recipe, and benchmark released; runs on stock vLLM with measured serving behavior (§10).

2. **A causal label recipe for the turn decision** (§4): one labeling rule and a minimal-pair construction, yielding 0.97 recall at 0.39 s P50 and 0.3 false fires/min with no inference-time crutches — confirmed on a fresh 2×-larger test set.

3. **A deployment-matched streaming benchmark** (§6): replayed real conversational timelines with a causally-decidable boundary taxonomy, under which the deployed VAD-timeout family is measured end-to-end — and under which our own historical model ranking *inverts*.
4. **Context-biasing numbers where none exist** (§8): +28.9 pp entity recall on Earnings-22, flat across session age; the fine-tune's offline WER cost measured *and attributed* by a marker-ablation.
5. **A transferable training lesson** (§5, §11): clairvoyant labels — how offline-derived supervision manufactures training oscillation and phantom trade-offs in streaming decisions, diagnosed with intervention experiments and repaired by relabeling alone.

2 Why a silence threshold cannot win

Two turns no timeout can get right

1. The pause that means "wait." *"My number is four one five . . . (0.9 s) . . . five five five . . . (1.1 s) . . . zero one nine two."* A 0.7 s timeout fires twice mid-number, and the agent answers with a partial phone number. The silence carries no evidence that the turn is over — but the *words* do: seven digits of a ten-digit pattern predict continuation. Production turn-detection teams report exactly this failure — a text-blind endpointer "detects the end prematurely after each digit" — and fixing it is worth double-digit reductions in interruptions [LiveKit, 2026].

2. The completion that means "go." *"Hello? I'm still here."* The thought is complete the instant the last word ends; a human interlocutor would respond within ∼200 ms [Stivers et al., 2009]. A 1.0 s timeout makes the user wait five times that long — for nothing. The semantics resolved the question before the silence started.

The two examples fail in opposite directions, and that is the point: the first needs the system to hold *through* long silence, the second to fire at *near-zero* silence. A single threshold cannot do both, and the gap between the two cases is not measurable in seconds of quiet. (Both examples are measured, not rhetorical: §9 builds probes for exactly these scenarios and closes the dictation gap with training data alone.) On our own conversational benchmark the silence-gap distributions of within-turn pauses and between-turn boundaries overlap almost completely (medians 1.39 s vs 0.83 s — the *pauses* are longer; §5); Skantze [2021] documents the same overlap across the dialogue literature, together with the resulting no-good-threshold dilemma that has defined cascade endpointing since Raux and Eskenazi [2008]. The resolving information is semantic completeness, which is exactly what a recognizer computes on the way to a transcript. That is the case for making the recognizer itself turn-aware — the decision rides on representations the ASR model already has, at zero marginal latency, rather than on a second model downstream of finalized text.

Transcription has its own version of the same argument. A voice agent hears names, tickers, account emails, drug names, part numbers — tokens whose correct spelling is nearly unrecoverable from acoustics alone but obvious given *who is calling about what*. Every major commercial STT ships a context mechanism (keyterm prompting, word boost, phrase lists), yet none publishes controlled uplift numbers, and the open model we build on describes its context facility in one qualitative sentence [Shi et al., 2026]. A voice agent has this context —

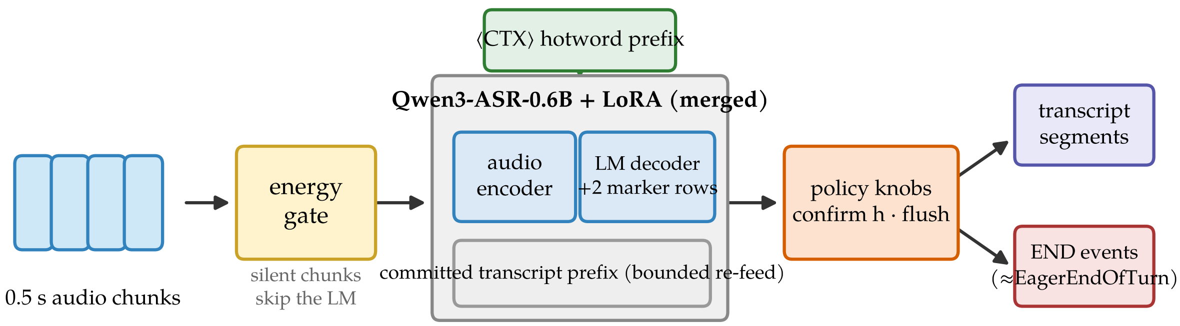

Figure 2. System shape. Audio chunks pass an RMS energy gate (silent chunks never start a segment, so mostly-silent sessions are nearly free); the merged LoRA model transcribes the current bounded segment with the `<CTX>` hotword prefix in the system slot and may emit marker tokens; two policy knobs (confirm horizon, max-segment force-flush) sit *behind* the model. Outputs are transcript segments plus END events — the same event shape as Flux's `EagerEndOfTurn`.

the user's profile, the account's entities — at session start; a turn-aware model with a context slot can ground the whole session's transcription in it. We measure exactly this in §8, and a concurrent benchmark confirms the effect across commercial systems [Durmus et al., 2026].

3 The system

Task. A single-channel audio stream arrives in 0.5 s chunks. At each chunk boundary the system must (a) extend a running transcript and (b) decide whether the turn has ended — an irreversible, latency-sensitive event that downstream components act on. A text context prefix (user profile, hotword list) may be supplied at session start and should bias recognition throughout.

Model. We start from Qwen3-ASR-0.6B [Shi et al., 2026], an open unified speech model (AuT audio encoder + dense Qwen3 decoder) with strong offline WER but no endpointing and no published context-biasing numbers. We claim two reserved vocabulary slots as marker tokens, `<EAGER_END_SPEECH>` and `<END_SPEECH>`, and LoRA-fine-tune [Hu et al., 2022] the decoder (r=16, α=32, q/k/v/o projections, plus the two marker embedding rows) to emit them inside the transcript at endpoint moments. Training data is $\sim$12k examples synthesized from LibriSpeech [Panayotov et al., 2015] and AMI [Carletta et al., 2005] under the label rule of §4; training takes hours on one GPU.

Inference policy. Three orthogonal, interpretable knobs (Figure 2); every reported arm states its knob settings. The **energy gate** never *starts* a segment on a silent chunk — motivated by a data blind spot (§6), and also the serving trick that lets one engine carry many mostly-silent sessions (§10). The **confirm horizon** h holds a fire candidate for h further silent chunks before signaling END (resumed speech cancels the signal; the transcript flush stands) — the phrase-vs-turn dial, which the final model barely needs. The **max-segment force-flush** bounds the committed transcript segment; a transcription reset, never an END event.

Table 1. The landscape (verified July 2026; details in §12). "Turn detectors" = Smart Turn (acoustic), LiveKit and TEN (text/audio EOU): separate models beside a recognizer. "Cascade" = VAD + silence timeout + separate ASR, the deployed default and our measured baseline in Figure 1. Kyutai STT's semantic-VAD head and Parakeet's `<EOU>` token are open-weights capabilities without published training accounts; Smart Turn is the one fully open recipe, but it does not transcribe.

	This work	Flux / `sem.vad`	Kyutai STT	Parakeet-EOU	Turn det.	Cascade
Streaming ASR	✓	✓	✓	✓	—	✓
In-model end-of-turn	semantic	semantic	sem. head	token	sep. model	timeout
Open weights	✓	—	✓	✓	mostly	✓
Turn-training recipe	✓	—	—	—	Smart Turn	n/a
Context biasing (numbers)	✓ (+28.9 pp)	—	—	—	—	ASR-dep.

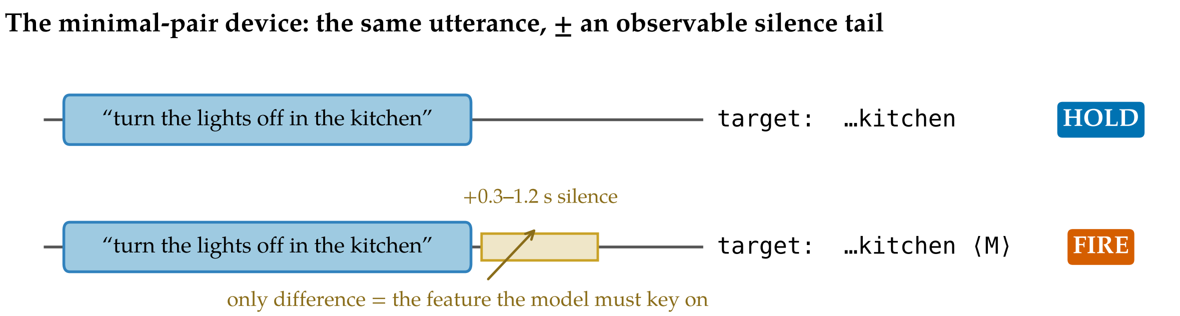

Figure 3. The minimal-pair device. Each complete utterance appears twice in the training data: without a silence tail (target: transcript only) and with a 0.3–1.2 s tail (target: transcript + marker). The only difference between the opposite-labeled examples is the observable the model must key on. This construction is what removes the training pathology of Figure 4.

4 The training recipe: one causal rule

The entire turn capability comes from the labels. One rule generates every training example: *emit `<EAGER_END_SPEECH><END_SPEECH>` at a point iff the speech so far is semantically complete AND $\geq$0.3 s of silence has elapsed since the last speech.* Both conditions are functions of the audio prefix — a property that turns out to be the load-bearing design decision (§5). Completeness is decided by a transcript heuristic (trailing filler/connective, mid-word cut, or very short non-acknowledgment ⇒ incomplete; Appendix A); silence is decided by the synthesized waveform itself.

Two constructions carry the semantics-vs-silence discrimination that motivated §2. The **minimal pair** (schemas 1–2, Figure 3) presents the same complete utterance with and without an observable silence tail, with opposite targets: completeness alone must not fire — the model has to *see* the silence. The **pause pair** (schemas 4–5) presents two-utterance sequences whose gap is bridged or fired depending only on whether the words before the pause are complete: ". . . and then, *(pause)*" holds; ". . . that's everything. *(pause)*" fires, even if the speaker later resumes. Quick resumes after a legitimate fire become two transcript segments — a measured cost, not an error (§6) — because at decision time the correct answer genuinely does not exist on the channel. Pair examples use both same- and different-speaker sources: the fire decision keys on completeness plus silence, never on speaker identity, which single-channel deployment cannot observe.

The recipe holds architecture, adapter, data volume, and source corpora identical to eight

Table 2. The eight training schemas (∼12k examples; LibriSpeech and AMI roughly 50/50). M = <EAGER_END_SPEECH><END_SPEECH>. Schemas 1–2 are the minimal-pair device (Figure 3); 4–5 teach pause discrimination from the *prefix* (hold only when the words so far are incomplete); 7–8 cover the silence-heavy channel shape that pure-utterance corpora never show (§6).

#	Construction	Target
1	complete utterance + 0.3–1.2 s silence tail	`text` M
2	*same utterances as #1*, tail $\leq$0.1 s	`text`
3	speech truncated mid-utterance at 30–80%	partial `text`
4	complete A + gap 0.3–2.5 s + B (+ tail)	`A` M `B` M / `A` M `B`
5	*incomplete* A + gap + B + tail	`A B` M
6	complete utterance + long silence (2–4 s)	`text` M
7	pure silence 0.5–4 s	(empty)
8	leading silence 0.5–2 s + utterance + tail	`text` M

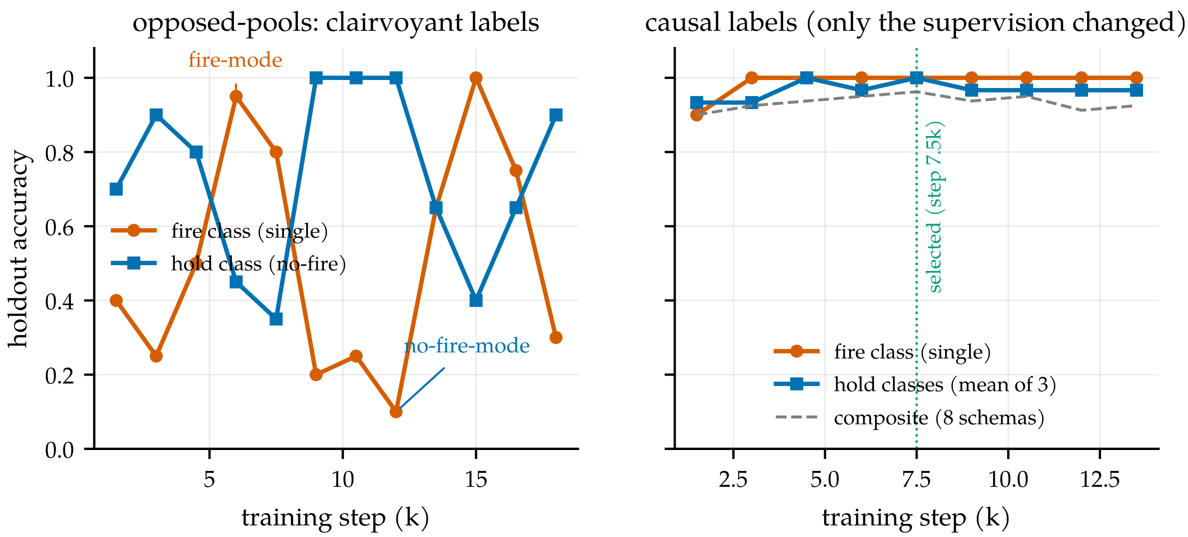

Figure 4. The fingerprint of contradictory supervision. Left: under the opposed-pools recipe (clairvoyant labels), holdout accuracy oscillates for the entire run between a *fire-mode* attractor (fire class ≈1.0, hold class ≈0.4) and a *no-fire-mode* attractor (fire ≈0.1, hold ≈1.0): the pools assign opposite targets to causally identical audio, so no single policy satisfies both. Right: the causal recipe of §4 — same architecture, same LoRA recipe, same data volume, *only the labels changed* — climbs monotonically, and both classes sit at ceiling *simultaneously* from step 3k on.

failed predecessors; the label rule is the only change. What the predecessors did wrong — and what their failure looks like from the outside, so the reader can recognize it in their own pipeline — is the subject of the next section.

5 What went wrong eight times: clairvoyant labels

Eight supervision recipes preceded §4, all on the same architecture, and their record reads in hindsight like a controlled study of what contradictory supervision does to a model. Checkpoints that fired promptly at turn ends also fired spuriously mid-utterance; checkpoints that never fired spuriously would not fire at all; late recipes oscillated between the two behaviors for entire training runs (Figure 4, left). The record at the eighth recipe concluded the behaviors

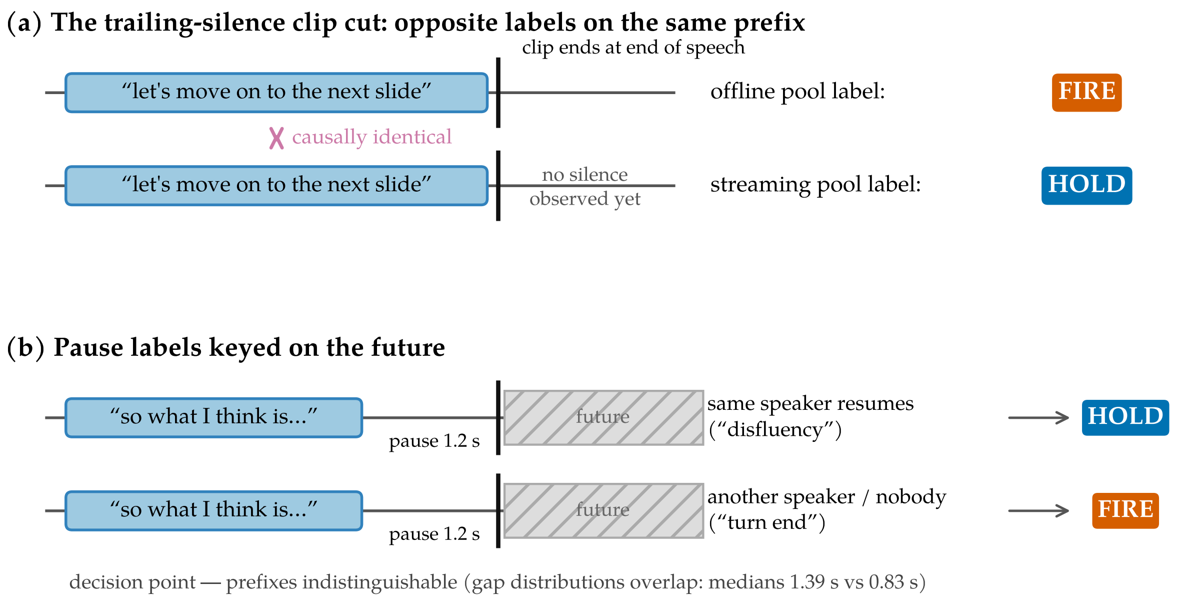

Figure 5. Two ways offline-derived supervision demands clairvoyance. (a) Offline clips inherit forced-alignment cuts at the end of speech, so the offline pool demands a fire on exactly the prefix the streaming pool labels *hold* — and the condition "speech ended, no silence yet" is one deployment never presents, because silence keeps arriving after a real speaker stops. (b) Disfluency-vs-boundary pause labels are keyed on who speaks *next*; at the decision point the prefixes are statistically indistinguishable (the two classes' silence-gap distributions overlap almost completely).

lay on a fundamental Pareto frontier and recommended shipping two checkpoints, one eager and one conservative. The frontier was manufactured — by the labels — and the mechanism is worth naming precisely, because nothing about it is specific to speech:

Clairvoyant labels

A label for a streaming decision at time t is *clairvoyant* if its value depends on input after t. Pools containing clairvoyant labels do not merely add noise: because the dependence is systematic, they partition the data into internally consistent subsets with incompatible optima. The observable symptoms are (a) training-time oscillation between mode attractors and (b) apparent capability trade-offs across checkpoints — a phantom Pareto frontier. **Principle: every training label and every evaluation target for a streaming decision must be computable from the causal prefix.**

Mechanism 1: the trailing-silence clip cut. Offline ASR clips end at the end of speech — datasets are segmented by forced alignment, and evaluation clips inherit the cut. An offline endpoint benchmark built on such clips therefore demands a fire on audio that ends during or immediately after speech. The streaming pools (correctly) teach the opposite: never fire before silence is observed. The condition "complete utterance, no trailing silence" thus carries the label *fire* in one pool and *hold* in the other (Figure 5a). In the eighth recipe the two pools were disjoint sets of AMI utterances with opposite labels on identically distributed content: 50/50 label noise on the exact decision the model exists to make.

Table 3. The supervision-composition record. Eight recipes, one architecture; each row adds one labeled pool (cumulative). "Early" latencies are negative: the model fires seconds before the turn ends. Every fix moved the model *along* the apparent frontier, never off it — until the labels themselves changed.

#	Supervision change (cumulative)	Observed behavior
1	offline endpoint pool only: fire at every clip end	fires eagerly and "accurately" offline (100% fire recall); in streaming replay, fires 89× per speech-minute mid-utterance
2	+ truncated-speech hold pool (11%→38% of mix)	diminishing returns: median fire timing improves only $-13.1 \to -9.6 \to -7.6$ s (still seconds early) as the dose rises
3	+ trailing-silence fire pool	streaming latency crosses zero (P50 +0.32 s) *but* offline fire recall regresses 100→82%; the "trade-off" is first declared
4	+ complete utterances with *and* without silence tails, both labeled fire	silence becomes optional again; a strictly intermediate point on the same frontier; frontier declared fundamental
5	+ hold pool on the *same* audio as existing fire examples	collapse: opposite labels on identical inputs; the model stops firing on meeting audio entirely (fire recall 0%)
6	+ hold pool on *disjoint but identically distributed* audio	bimodal oscillation between fire-mode and no-fire-mode attractors for the whole run (Figure 4, left); no checkpoint dominates; postmortem recommends shipping two checkpoints
7	**causal relabel (§4):** fire iff complete *and* ≥0.3 s observed silence	monotone training, both classes at ceiling simultaneously; single checkpoint dominates every predecessor (§7)

Mechanism 2: labels keyed on who speaks next. The historical pools separated "disfluent pause — hold" from "turn boundary — fire" using the transcript's segmentation: same speaker continues ⇒ disfluency; another speaker follows ⇒ boundary. Measured on the fifth recipe's pools, the two classes' silence-gap distributions overlap almost completely (medians 1.39 s vs 0.83 s). At decision time — mid-pause — the classes are indistinguishable; the label is a function of the future (Figure 5b). An offline decoder can satisfy such labels by peeking; a causal model cannot, and part of the historical "disfluency over-fire" penalty punished models for lacking clairvoyance.

The record, and the intervention that falsified the frontier. Table 3 compresses the eight recipes into a dose–response record: every added pool moved the model along the apparent recall-vs-precision frontier, never off it. The frontier's fire-recall axis died by a one-variable intervention on the *evaluation*: appending 1.0 s of silence to the offline clips — restoring the future that deployment always delivers — took the "broken" eighth recipe's fire recall from 0.10 to 1.00 and the fifth's from 0.82 to 0.98, while the offline-labeled first recipe, which never depended on hearing silence, was the unchanged control (Table 4). Under deployment-realistic conditions all three fire essentially perfectly on complete utterances: the axis four data recipes had been engineered against was an artifact of where the clips ended.

Table 4. The silence-append intervention. Fire recall on complete single utterances from the legacy offline benchmark, before and after appending 1.0 s of silence to each clip.

Model (supervision recipe)	original clips	+1.0 s silence	Δ
offline-labeled (rows 1–2)	1.00	1.00	0.00
mixed-pools (row 3)	0.82	0.98	+0.16
opposed-pools (row 6)	0.10	1.00	+0.90

How a contradiction expresses itself. Rows 5 and 6 of Table 3 form a natural two-condition experiment. When opposite labels sit on literally identical audio (row 5), the conflict is visible to the loss on every example and the model settles deterministically on one side — it simply stops firing. When opposite labels sit on disjoint but identically distributed audio (row 6), no single example exposes the conflict; each pool is separately learnable, incompatible only in aggregate, and training *circulates* between mode attractors that each satisfy one pool. The oscillation is not instability to be scheduled away; it is the loss surface reporting a contradiction that no example states outright. The causal relabel (row 7) removes both fingerprints at once — Figure 4 (right): the composite climbs monotonically 0.900→0.963, and fire and hold classes sit at ceiling simultaneously for the entire run.

6 Measuring what deployment sees

If clairvoyant labels are the disease, evaluation is where it enters — our legacy offline benchmark did not just mis-score the models above, it drove four recipes of wasted data engineering. We replaced it with one deployment-matched protocol, and applied the causality principle to the benchmark itself.

Material and ground truth. Continuous single-channel stretches from held-out AMI meetings: one target speaker's utterances at their true timeline offsets, silence between them, 30–60 s per stretch (the voice-assistant shape: one microphone, one user; audio never ends at a speech boundary). Each utterance-end boundary is classified by what is observable *on this channel*: **turn-final** (same-speaker gap ≥ 2 s) — should fire; **continuation** (gap in $[0.3, 2)$ s, same speaker resumes) — firing is a measured segmentation *cost* (`resume_after_fire`), not an error, because the correct answer depends on the future; **internal** (pauses inside utterances, or <0.3 s after speech) — firing is a *false fire*. Metrics: boundary recall within $[-0.25, +1.5]$ s, false fires per speech-minute, latency percentiles, resume rate, streaming WER of flushed transcripts, per-chunk compute. An early version of the benchmark itself contained a clairvoyant rule (promoting boundaries to turn-final when *another meeting speaker* interleaved — inaudible on this channel); our own checklist (§11) caught it, and all numbers use the corrected classification (Appendix B).

Finding 1: silence incompetence. Figure 6. Every pre-diagnosis training example begins with speech; a real channel is mostly silence. On silence the models hallucinate and fire; ungated numbers are uninterpretable, and all comparisons below run gated.

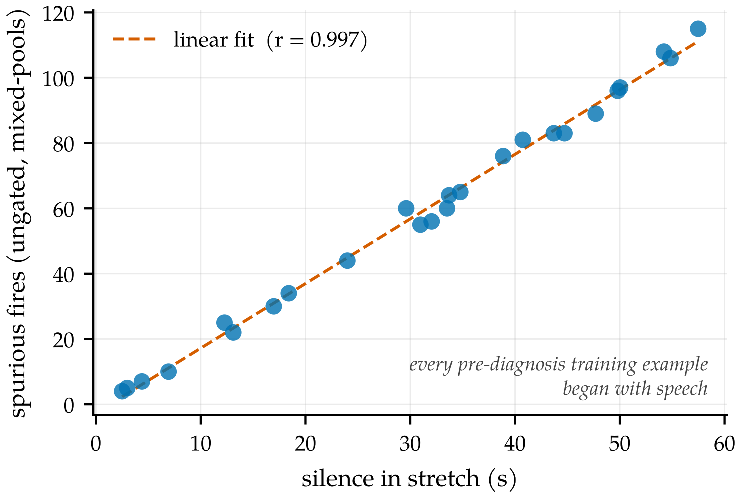

Figure 6. Silence incompetence. Spurious fires of the ungated mixed-pools model against the silence content of each benchmark stretch: the model hallucinates text and sprays markers on silence at ~1.9 fires/s ($r = 0.997$), because every pre-diagnosis training example begins with speech while a real channel is mostly silence (13 of 22 minutes here). Consequence: *ungated* recall is uninterpretable — a random 1.9 Hz spammer scores 0.96 recall against the tolerance window. The fix costs one line (RMS energy gate), and the causal recipe additionally bakes silence schemas into training (Table 2, #7–8).

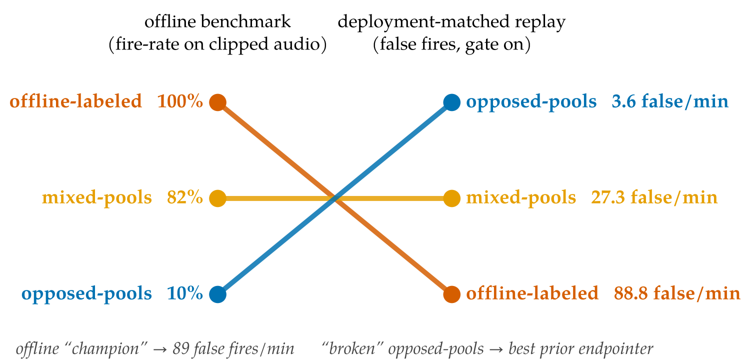

Figure 7. The ranking inversion. Left: the legacy offline benchmark (fire-rate on clips cut at end-of-speech). Right: the deployment-matched replay benchmark (false fires per speech-minute, energy gate on; full metrics in Table 5). The offline champion fires 89 times per speech-minute mid-utterance — its offline strength *is* a premature-fire pathology once audio keeps flowing. The opposed-pools model, written off at 10% offline, is the best prior endpointer. Both historical ship decisions were wrong.

Finding 2: the ranking inverts. Figure 7. A benchmark with clairvoyant targets does not just mis-score models; it mis-*ranks* them, and the error is largest exactly at the top — the offline-labeled model was "best" offline *because* it fires without waiting for silence, which is the single worst property in deployment. Any team selecting a turn-aware checkpoint on clip-cut offline evals should expect the same inversion.

7 Endpointing results

Dominance, and escaping the curve. Table 5 and Figure 1 (page 3). Against its predecessors, the causal model with the gate alone is better on every axis at once. Against the deployed

Table 5. Main result (deployment-matched replay, energy gate on; latency-vs-false-fire view in Figure 1). Rows 1–4: the 25-stretch development benchmark (96 turn-final boundaries). The causal model dominates every predecessor without either crutch: its *raw* false-fire rate beats the mixed-pools model's *confirmed* rate, and its WER needs no force-flush. The confirm ablation: the mixed-pools horizon cancels 241 phrase-boundary candidates; the causal model's cancels 3. Last row: a *fresh* 2×-larger stretch set (50 stretches, 184 boundaries) drawn after all development ended — latency and false fires hold; recall dips 4.5 pp (§13). †WER mean dominated by long-monologue repetition loops when no force-flush bounds the segment (causal fresh-set WER *median* 0.30, matching the development set; the §10 force-flush bounds this tail).

Policy	Recall ↑	P50 ↓	P95 ↓	False/min ↓	WER_{mean} ↓
mixed-pools + gate + confirm $h=1$	0.927	0.68 s	1.27 s	3.2	0.36
opposed-pools + gate + confirm $h=1$	0.938	0.77 s	1.06 s	0.0	4.96†
causal + gate (ours)	**0.969**	**0.39 s**	**0.70 s**	**0.3**	**1.43**
causal + gate + confirm $h=1$	0.969	0.89 s	1.20 s	0.0	1.43
causal + gate — fresh 50-stretch set	0.924	0.42 s	0.70 s	0.2	4.63†

cascade, the comparison's *structure* is the point: a timeout must be long enough to survive within-turn pauses, so it can only trade latency against false fires along one curve; the causal model escapes the curve by reading completeness — it fires 0.39 s after a finished thought and holds through a 1.4 s mid-sentence pause, which is exactly the pair of behaviors §2 argued no threshold can combine. The inference crutches retire: the confirm horizon that earlier recipes needed to be usable cancels only 3 candidates for the causal model (vs 241 for the mixed-pools model) — the phrase-vs-turn discrimination moved from inference policy into weights, which is what supervision was supposed to do all along. One honest caveat: quick resumes after a legitimate fire are always segmented (resume rate 1.0, by design — the correct choice is unknowable at decision time); deployments preferring merged segments pay +0.5 s via confirm $h=1$.

Held-out confirmation and uncertainty. The 25-stretch development benchmark was reused across development, so we drew a fresh 2×-larger set (50 stretches, 184 turn-final boundaries) after all development ended. Latency, false fires, and WER median reproduce; recall drops 0.969 → 0.924. The 95% Wilson intervals — $[0.91, 0.99]$ on the development set (93/96) and $[0.88, 0.95]$ on the fresh set (170/184) — overlap, so the drop is within sampling variation, though its direction is consistent with mild adaptive overfitting to the development set. We treat the fresh set as the honest operating estimate. Read against the same-set timeout family (Appendix C): the cascade can reach higher recall on this set (0.978–0.989 at $X \in \{0.5, 1.0\}$ s) but only at 1.7–2.9× the causal model's latency and 3–22× its false fires — and no timeout setting reaches the causal model's (0.42 s, 0.21/min) corner at *any* recall.

8 Context-biased transcription

The second half of the system claim is that one model can also ground its transcription in session context — and that an endpoint fine-tune does not wreck the recognizer it lives in.

Context biasing, measured where it matters. Figure 8(a): on Earnings-22 [Del Rio et al., 2022] — spontaneous earnings-call speech dense in tickers, executives, products; entities

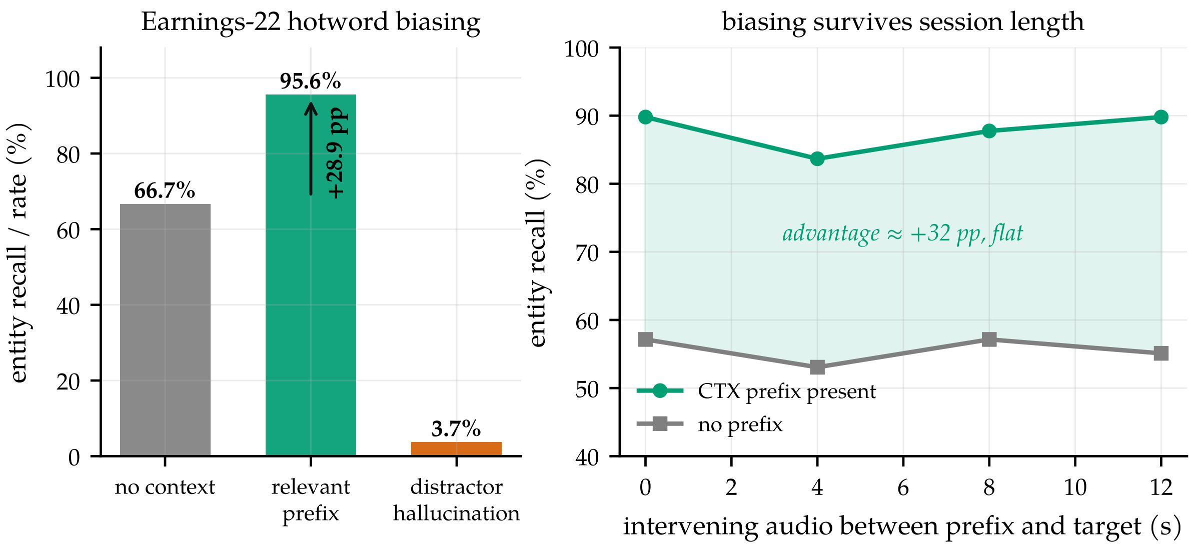

Figure 8. Context biasing: the capability with no public numbers. (a) Earnings-22 entity recall without context, with the relevant hotword prefix, and the hallucination rate under a distractor prefix (150 utterances; LLM-extracted proper-noun entities). (b) Recall vs seconds of unrelated speech interposed between prefix and target: the ≈+30 pp advantage does not decay with session age.

LLM-extracted and filtered to genuine proper nouns — a relevant hotword prefix lifts entity recall from 66.7% to 95.6% (+28.9 pp), while nudging WER down (15.3→15.0%): the prefix acts as a domain hint, not a distortion. A *distractor* prefix (another utterance's entities) induces 3.7% hallucination — above our 3% target, reported as-is; distractor-ratio training is the known, unexercised lever. Qwen3-ASR describes context biasing in one qualitative sentence and reports no numbers [Shi et al., 2026]; to our knowledge these are the first on this model family. A concurrent benchmark [Durmus et al., 2026] standardizes contextual evaluation on the same corpus and finds every commercial system tested gains keyword F-score from context — corroborating the mechanism's value and the field's shortage of controlled numbers.

The advantage survives session length. Figure 8(b): with 0–12 s of unrelated speech interposed between the prefix and the target words, the biasing advantage stays flat at ≈+30 pp. This is the property a voice agent actually needs — the user's account entities help at minute five, not just in the first utterance — and it is what makes the `<CTX>` slot a *session* capability rather than a per-request trick.

Offline ASR cost — measured and attributed. Table 6: the endpoint fine-tune costs +1.1 pp WER on LibriSpeech test-clean and +2.7 pp on test-other. Our first hypothesis was mechanical: the model fires its marker on 4.7–5.9% of offline clips (many LibriSpeech clips carry the ≥0.3 s silence tail where the causal rule *says* fire), and a fire ends the decode, truncating the transcript. We tested it by banning the two marker tokens at decode time — and the hypothesis *fails*: fires drop to zero, WER does not move (3.94%/7.91%). The regression is genuine distributional drift from fine-tuning on ∼12k narrow-schema examples with no plain-ASR replay data, and the standard mitigation (mixing transcription-only data into the fine-tune) is unexercised in this release. The deployment-relevant *streaming* WER is unaffected (Table 5, median 0.28–0.30 across sets), so the cost is confined to using the endpoint checkpoint as an offline transcriber —

a job the base checkpoint still does.

Table 6. Offline WER regression (LibriSpeech, full official splits, single-shot decode on vLLM, jiwer/whisper normalization). "Fires" = fraction of clips where the model emitted the endpoint marker (offline clips end at or near end-of-speech; base never fires because its marker rows are untrained).

	WER ↓		Fires	
Model	clean	other	clean	other
Qwen3-ASR-0.6B (base)	2.79%	5.14%	0%	0%
+ endpoint LoRA (merged, causal)	3.93%	7.87%	4.7%	5.9%
+ markers banned at decode	3.94%	7.91%	0%	0%

9 Closing the loop on §2: one model that also dictates

The two motivating scenarios deserve measurement, not anecdote, so we built probes for both and used them to drive the recipe to a single model that handles conversation, dictation, and context together. The **dictation probe** replays 50 ten-digit phone numbers (3-3-4 groups from held-out Free Spoken Digit Dataset speakers, 0.6–1.2 s inter-group pauses) through the exact streaming stack of §6; the **spelled-entity probe** decodes 80 synthesized utterances that spell uncommon names and email addresses, with and without a user-profile context prefix, plus a wrong-user *distractor* profile. Probe speakers, voices, and identities are all disjoint from training.

The conversational model fails the dictation probe — as predicted. Its completeness heuristic reads a digit group as a finished thought: 4.8 premature fires per dictated number, every sequence interrupted — *worse than a 0.5 s timeout* (2.0). Spelled emails are unusable: 0% exact without context, 40% with the profile. The §2 examples are real failure modes of the system as shipped in §7.

Dictation schemas close the endpointing gap. We extend the recipe with dictation and context schemas, built under the same causal discipline: partial phone numbers (any 1–9 digit prefix) followed by a pause are labeled *hold* — the pattern predicts continuation; full ten-digit numbers with the same internal pauses fire after the usual $\geq$0.3 s tail (and, without a tail, do not); spelled names and emails are targeted in normalized written form (`anna.kowalski@pine.ai`), half of them with a user profile in the context slot. Retraining the same LoRA on the extended pool ($\sim$14.7k examples, hours on one GPU) drives premature fires from 4.8 to 0.24 per number — better than *every* timeout setting — at 0.996 digit accuracy, and lifts spelled names to 100% exact and emails from unusable to context-grounded (Table 7).

The intrusion trap, and the causal fix. Teaching the model to *use* the profile first taught it to *copy* the profile: trained only on matching profiles, an early extension emitted whatever entity the context named, so a wrong-user profile wrote the wrong email into the transcript — 11.3% wrong-profile intrusion, rising to 40% with longer training. The fix is a labeling one, in the same spirit as the rest of the paper: a third of the spelled examples carry a *conflicting* profile — the context names one identity, the audio spells another, and the target follows the audio.

Table 7. Dictation and spelled-entity probes. Premature end-of-turn fires per dictated 10-digit number, final-boundary recall (window +1.75 s), digit accuracy; exact-match on spelled names/emails without/with the user-profile context; and the wrong-profile intrusion rate. Adding dictation schemas beats every timeout on premature fires but, trained on matching profiles only, copies a wrong profile 11.3% of the time; distractor-ratio training (context and audio disagree, target follows audio) removes that at no cost — one unified model. Residual dictation misses are ∼5 fires landing ∼2 s after speech end, just outside the window.

	Premat./seq ↓	Final rec. ↑	Digit acc ↑	Name −/+	Email −/+	Intrus. ↓
conversational only	4.80	0.76	0.910	0.30 / 0.83	0.00 / 0.40	1.3%
timeout X=0.5 s	2.00	1.00	—	—	—	—
timeout X=1.0 s	0.74	1.00	—	—	—	—
+ dictation (match-ctx)	0.36	0.78	0.996	1.00 / 1.00	0.03 / 0.93	11.3%
+ distractor (unified)	**0.24**	**0.80**	**0.99**	**1.00 / 1.00**	0.03 / 0.82	**0.0%**

This makes context a prior rather than a substitute for listening, and it is causally clean (the correct output is a function of the audio, never of the context). It drives intrusion to *zero at every checkpoint of the run* (from 40% to 0.000; Table 7), while leaving dictation and matching-context accuracy intact.

One model, four behaviors. The result is a single checkpoint that endpoints conversation, holds through dictation, grounds spelled entities in context, and ignores a wrong profile — no deployment-time checkpoint swap. On the conversational benchmark of §7 the unified model scores 0.938 recall at 0.65 false fires per speech-minute (within the Wilson interval of the conversation-only 0.969; §7), while adding 0.24-premature-fire dictation, +0.82 email-with-context, and 0.0% intrusion — the whole capability set at one operating point. Its offline WER (3.9%/7.3%) carries the same fine-tune regression as the conversation-only model; a controlled comparison in Appendix D shows the residual is narrow-schema drift, not an artifact of the training recipe, and identifies the standard mitigation (plain-ASR replay data) as the one unexercised lever. Appendix D also documents the AdamW hygiene we hardened along the way — a real weight-decay bug that silently shrinks the base model — and a negative result: the Muon optimizer underperforms AdamW on these LoRA factors.

10 Deployment reality check

Streaming papers routinely stop at a simulator. Three findings from serving this model on stock vLLM [Kwon et al., 2023] (one *shared* GPU throughout):

An OOD landmine. Figure 9(a). The natural flat-cost serving shape — append each 0.5 s chunk as a new audio placeholder in a growing resident request — destroys the model: 88% WER vs 3.8% for re-feeding the last window as a single segment. The base model was trained on single audio segments; per-chunk placeholders are wildly out of distribution. Bounded re-feed is the only correct primitive for this family.

The simulator was the bottleneck. Figure 9(b). The replay protocol on vLLM reproduces the endpoint behavior at 84 ms median per chunk vs the transformers simulator's 455 ms (5.4×) —

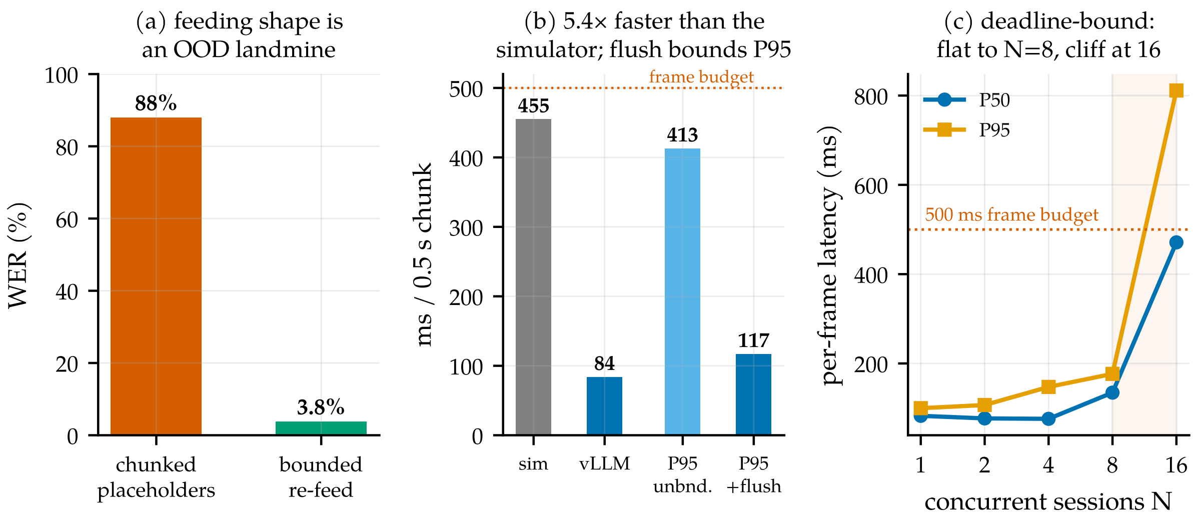

Figure 9. Serving on real infrastructure. (a) Feeding shape: per-chunk audio placeholders are out-of-distribution for a model trained on single segments (88% WER); bounded re-feed of the last window costs a re-encode but is correct (3.8%). (b) Per-chunk compute: the vLLM path is 5.4× faster than the transformers simulator used for all pre-serving latency numbers; the max-segment force-flush also bounds the P95 tail. (c) Concurrent sessions per engine on a 0.15-GPU slice: flat to N=8, cliff at 16 — the deadline binds, not memory.

∼6× real time on a 0.15-utilization slice. Latency claims made on $O(T^2)$ re-decode simulators overstate the cost of this approach by that factor.

Bound every resident state. Unbounded committed prefixes eventually loop on long monologues; a 20 s force-flush fixes the WER tail (2.54→1.29) *and* the compute tail (P95 413→117 ms) at no endpoint cost (Figure 9(b)). Under concurrency (Figure 9(c)) per-frame p95 stays flat through N=8 sessions per 0.15-GPU slice and cliffs at 16 — deadline-bound, not memory-bound, with the energy gate batching only the sessions actually due each frame. (A contention-caveated lower bound; the shape, not the constant, is the result.) The in-engine efficiency variant — windowed KV with the `<CTX>` prefix pinned as an attention sink [Xiao et al., 2024] — is validated separately in a companion systems paper [Li, 2026]; capability-wise, bounded re-feed already keeps biasing flat (Figure 8(b)).

11 Lessons for streaming supervision: a checklist

The failure mode of §5 generalizes to any model that must act mid-stream on offline-labeled data. Four questions for a streaming supervision pipeline:

Is your supervision clairvoyant?

1. **Prefix test.** For each label at decision time t: is it computable from input up to t? *(Our pause labels were keyed on the next speaker.)*
2. **Twin test.** Can two causally identical prefixes receive different labels anywhere in the data? *(Same complete-utterance prefix: fire in one pool, hold in the other.)* Minimal pairs differing in an *observable* are the fix, not the bug.
3. **Phantom-condition test.** Does the eval demand behavior on conditions deployment never produces? *(Clips cut at end-of-speech; deployment always delivers the next silence.)*
4. **Oscillation test.** Does training bounce between mode attractors that each satisfy part of the data? That is not instability to be scheduled away; it is the loss surface reporting a contradiction.

Candidates we believe are affected: turn-taking policies for full-duplex agents trained from offline dialogue transcripts (the labels encode who *did* speak next, not what was knowable — cf. the projection-style objectives of Ekstedt and Skantze, 2022, which are explicitly predictive and thus honest about the uncertainty); barge-in and endpointing in commercial stacks; simultaneous translation trained with reference-aligned read/write decisions [Ma et al., 2019]; streaming TTS interruption; tool-call timing for agents cloned from completed trajectories — a streaming sibling of causal confusion in imitation learning [de Haan et al., 2019]. In each, an apparent aggressiveness-vs-caution frontier may partly be an artifact of asking the model to know the future. The diagnosis also joins a broader lineage of measurement-induced phenomena: as sharp metric choices can manufacture "emergence" [Schaeffer et al., 2023], non-causal label choices can manufacture *trade-offs*.

12 Related work

Endpointing and end-of-turn detection. Classical endpointing thresholds a VAD [Silero Team, 2021] with a silence timeout, and threshold optimization has been studied since Raux and Eskenazi [2008]; learned refinements predict end-of-query directly from acoustics [Shannon et al., 2017], jointly train an endpointer with the recognizer [Chang et al., 2019, Li et al., 2020, Bijwadia et al., 2023], or distill tiny acoustic end-of-turn models [Helwani et al., 2026, Pipecat team, 2025]. Recent systems fuse acoustic and linguistic cues for the turn decision [Wang et al., 2026, Yang et al., 2026] or fold VAD, turn detection, and ASR into one audio-LLM front-end [Li et al., 2026] — the latter architecturally closest to us, but with no released weights or recipe. Deployed open detectors run *beside* the recognizer: Smart Turn (acoustic, fully open recipe, no transcript; Pipecat team, 2025), LiveKit's turn detector and TEN (end-of-utterance classifiers over a separate STT's output; LiveKit, 2025, TEN Team, 2025, LiveKit, 2026). Open-weights streaming recognizers with in-model turn signals exist without training accounts: Kyutai STT's semantic-VAD head [Kyutai Labs, 2025] and NVIDIA's Parakeet-EOU token [NVIDIA, 2026] publish weights but neither the labeling procedure nor the data construction for the turn decision, and neither measures context biasing. Commercially, Deepgram Flux integrates end-of-turn events into the recognizer's forward pass [Deepgram, 2026], OpenAI exposes

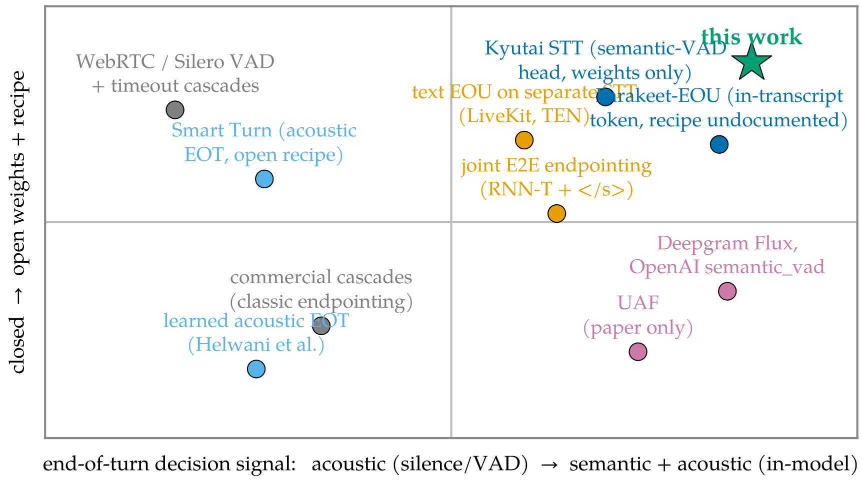

Figure 10. Where this work sits (verified July 2026). End-of-turn systems by decision signal (acoustic silence vs. in-model semantics) and openness (weights *and* recipe). The semantic in-model quadrant's open corner is thinly populated: Kyutai STT ships a semantic-VAD head and Parakeet-EOU an in-transcript token, both without published training accounts; Smart Turn opens a full recipe but is a detector, not a recognizer; LiveKit and TEN classify downstream of a separate STT. This work contributes the missing combination: in-transcript semantic end-of-turn with open weights, open recipe, and measured context biasing.

semantic VAD [OpenAI, 2025], and AssemblyAI ships "intelligent endpointing" [AssemblyAI, 2025]; none publishes training details. Our contribution to this line is the missing artifact — the recipe — plus evidence that the recall-vs-precision trade-offs this literature often reports as intrinsic can be manufactured by non-causal supervision (§5).

Turn-taking in dialogue. Turn-taking prediction has a long tradition in conversational AI [Skantze, 2021], from text-based end-of-turn prediction [Ekstedt and Skantze, 2020] to voice-activity projection trained on future windows [Ekstedt and Skantze, 2022]; the human timing evidence (200 ms median gaps, projection rather than reaction) comes from Stivers et al. [2009] and Levinson and Torreira [2015]. Projection objectives predict distributions over the future and are therefore causally honest; our diagnosis concerns the sharper failure where future-dependent quantities are presented as *deterministic labels*.

Streaming and unified ASR. Streaming recognition spans RNN-T [Graves, 2012], large weakly supervised models [Radford et al., 2023], decoder-only streaming with latency losses [Wan et al., 2026], and unified streaming/offline speech-LLMs [Shi et al., 2026]. Kyutai's delayed-streams models add a semantic-VAD head to streaming STT [Kyutai Labs, 2025, Défossez et al., 2024]; full-duplex speech-to-speech models [Défossez et al., 2024] dissolve the turn abstraction entirely and are a different product class. We inherit the unified-ASR substrate and add the in-transcript turn event plus the training recipe.

Context biasing. Contextual ASR ranges from attention-based deep context [Pundak et al., 2018] to retrieval-scale biasing [Gong et al., 2025], RL-tuned hotword integration [Kong et al., 2025], and cue-based prompting for speech LLMs [Novitasari et al., 2026]; Contextual Earnings-22 [Durmus et al., 2026] concurrently standardizes entity-level evaluation on the corpus we use.

We use plain prompt-slot biasing and contribute measurements on the open unified-model class: uplift on entity-dense audio and durability across session age.

Serving. vLLM's paged KV [Kwon et al., 2023] carries our serving path; attention sinks [Xiao et al., 2024] and bounded-state streaming serving [Li, 2026] provide the in-engine efficiency variant of our pinned context prefix.

13 Limitations

Single-speaker, single-channel English; the mixed-channel meeting shape is future work. The dictation gap of §2 is now measured and closed by a single unified model (§9), but the continuation-expecting structure it learns is the ten-digit phone shape; other enumerations (street addresses, serial numbers, card numbers) are untrained, a learned completeness judge remains the general lever, and $\sim$10% of dictation final fires land $\sim$2 s late. The unified checkpoint's conversational recall (0.938) is within sampling noise of the conversation-only model (0.969) but nominally lower; a deployer prioritizing raw turn-recall over dictation may still prefer the latter. Each supervision recipe in Table 3 was trained once; multi-seed replication of the oscillation-vs-monotone contrast, and a component ablation of the causal recipe, are the immediate next experiments. Distractor hallucination on Earnings-22 (3.7%) exceeds our 3% target. Quick resumes are always segmented (resume rate 1.0); merging them costs $+0.5$ s via the confirm knob or requires post-hoc fusion. Concurrency numbers are contention-caveated lower bounds from a shared GPU. The development benchmark was reused during development; the fresh-set numbers (recall 0.924, overlapping Wilson intervals, §7) are the honest operating estimate. The endpoint fine-tune costs offline WER (§8): mixing plain ASR data into the fine-tune is the standard, unexercised mitigation.

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A Label-specification details

Completeness classifier for pair examples (Table 2, #4–5): utterance A is *incomplete* iff its transcript ends in a filler/connective ("and, but, so, or, uh, um, the, a, to, of, in, with, i mean, you know, . . . "), ends mid-word (AMI partial-word annotation), or has fewer than 3 words outside a closed acknowledgment list. Borderline cases land in the `resume_after_fire` cost knob — they never create fire/hold contradictions on identical observables. Abandonment (incomplete speech, speaker never resumes) is an inference-policy timeout, not a label. Pair examples use both same- and different-speaker AMI pairs: the fire decision keys on completeness + silence, never on speaker identity, which single-channel deployment cannot observe.

B Benchmark specification and the corrected classifier

Full protocol: 0.5 s chunks; committed-prefix incremental decoding (emitted text committed, last 5 tokens rolled back) matching the base model's official streaming semantics; fires timestamped at the chunk boundary where `<END_SPEECH>` first appears; recall window $[-0.25, +1.5]$ s clipped at the next utterance's onset. The first version of the boundary classifier promoted a boundary to turn-final when another meeting speaker interleaved before the target speaker resumed. On a single-channel stream that speech is silence — demanding a fire there is exactly the clairvoyance the benchmark exists to remove. The corrected classification moved the mixed-pools model's recall 0.917→0.906 and the opposed-pools model's 0.935→0.938; no conclusion changed, but the episode is a working example of §11 applied to one's own evaluation.

C Additional results

Gated arms and the confirm-horizon sweep. Table 8. The confirm horizon behaves as a clean latency-vs-false-fire dial: h=1 strictly improves the mixed-pools model (recall *rises* because deferring the fire lets the segment gather the boundary inside tolerance) and zeroes the opposed-pools and causal models' false fires; h=2 over-suppresses (mixed-pools recall 0.812, latency past budget). The causal model is the only checkpoint whose h=0 row is already deployable. Ungated arms are omitted as uninterpretable (Figure 6): with ∼1.9 spurious fires per second on silence, the pre-diagnosis models score 0.95–0.99 "recall" by chance; their gated rows below are the meaningful ones.

The full timeout family. Table 9, both detectors on the development (25 stretches) and fresh confirmation (50 stretches) sets. The family's shape is stable across sets: recall collapses between X=1.0 and 1.5 s (the fire lands outside the +1.5 s tolerance), and no setting reaches the causal model's (recall, latency, false-fire) point. Silero ran per-chunk with state resets — a slight handicap; RMS shares the LM arms' exact detector and is the apples-to-apples policy comparison.

Fresh-set confirmation. The causal model with the gate on the confirmation set: recall 0.924, P50 0.42 s, P95 0.70 s, 0.21 false fires/min, resume 0.91, WER median 0.30 / mean 4.63 (three long-monologue stretches enter the no-flush repetition loop; the §10 force-flush is the

Table 8. All gated arms on the 25-stretch development benchmark. Resume = `resume_after_fire` rate over continuation boundaries (a cost knob, not an error rate).

Arm	Recall	P50	P95	False/min	Resume	$\mathrm{WER}_{\mathrm{mean}}$
offline-labeled + gate	0.896	0.05 s	0.25 s	88.8	0.89	0.85
mixed-pools + gate	0.906	0.16 s	0.89 s	27.3	0.96	0.36
mixed-pools + gate + confirm h=1	0.927	0.68 s	1.27 s	3.2	0.85	0.36
mixed-pools + gate + confirm h=2	0.812	1.15 s	1.42 s	1.1	0.56	0.36
opposed-pools + gate	0.938	0.26 s	0.56 s	3.6	0.93	4.96
opposed-pools + gate + confirm h=1	0.938	0.77 s	1.06 s	0.0	0.85	4.96
causal + gate	0.969	0.39 s	0.70 s	0.3	1.00	1.43
causal + gate + confirm h=1	0.969	0.89 s	1.20 s	0.0	0.93	1.43

Table 9. Silence-timeout cascade, all settings, on the development (dev) and fresh confirmation (conf) stretch sets.

Detector	Set	X (s)	Recall	P50	P95	False/min	Resume
RMS	dev	0.5	0.958	0.65 s	0.94 s	6.3	1.00
RMS	dev	1.0	0.969	1.15 s	1.44 s	1.4	0.81
RMS	dev	1.5	0.208	1.38 s	1.49 s	0.8	0.07
RMS	dev	2.0	0.031	0.55 s	0.55 s	0.2	0.00
RMS	conf	0.5	0.978	0.70 s	0.95 s	4.6	0.99
RMS	conf	1.0	0.989	1.20 s	1.45 s	0.6	0.73
RMS	conf	1.5	0.185	1.44 s	1.50 s	0.4	0.07
RMS	conf	2.0	0.016	1.14 s	1.14 s	0.2	0.00
Silero	dev	0.5	0.844	0.54 s	0.88 s	13.5	0.96
Silero	dev	1.0	0.854	1.04 s	1.38 s	5.5	0.85
Silero	dev	1.5	0.417	1.24 s	1.49 s	2.5	0.41
Silero	dev	2.0	0.125	1.32 s	1.42 s	1.0	0.00
Silero	conf	0.5	0.837	0.51 s	0.83 s	12.3	0.88
Silero	conf	1.0	0.864	1.00 s	1.33 s	4.4	0.73
Silero	conf	1.5	0.451	1.29 s	1.48 s	2.0	0.27
Silero	conf	2.0	0.130	1.34 s	1.44 s	1.3	0.07

documented fix). The 4.5 pp recall drop from the development set is the generalization gap discussed in §7 and §13.

D Training-recipe notes

The unified model (§9) uses the same LoRA fine-tune as the rest of the paper, plus two AdamW hygiene fixes and one negative result. All three were isolated on identical data (the unified pool), so the comparisons below vary only the optimizer setting.

A weight-decay bug worth fixing, but not the WER culprit. We reserve two vocabulary rows as markers and train them by unfreezing `embed_tokens/lm_head` and masking gradients to those two rows. But AdamW's *decoupled* weight decay applies $p \leftarrow p - \eta\lambda p$ to *every* row regardless of gradient, so at η=2×10^{-4}, λ=0.01 the entire (tied) 151k-row matrix shrinks ∼2.4% over 12k steps — silently degrading the base model's vocabulary and output head. This is a general hazard for any recipe that adds a few trainable vocabulary rows to a frozen embedding, and we fix it with a $\lambda=0$ group. *However*, honesty requires the negative half: in a controlled comparison on identical data, the fix (plus cosine schedule) did *not* lower offline WER — the fixed unified checkpoint scores 3.9%/7.3% (clean/other) versus 3.6%/7.0% without it, a difference within checkpoint-selection noise and in the wrong direction. The offline-WER regression of §8 is therefore *not* caused by the weight-decay bug; it is genuine narrow-schema drift from fine-tuning on $\sim$12k examples with no plain-ASR replay data. We keep the fix for its base-model-preservation rationale (a shrinking embedding is a latent hazard as training scales), not for a WER win we cannot demonstrate.

Cosine decay. A cosine schedule from 2×10^{-4} to 10^{-5} after a 100-step warmup gives the most stable, highest holdout-composite trajectory (plateau 0.992 from step 6k) and a slightly less trigger-happy final checkpoint (fewer false fires, lower offline marker-fire rate) than the constant schedule.

Muon underperforms AdamW here. We tried Muon [Jordan et al., 2024] on the 2-D LoRA factors (Newton-Schulz-orthogonalized momentum, AdamW retained for the marker rows) at a $100\times$ larger base LR (2×10^{-2}, since the orthogonalized update is $\sim$unit-norm). On identical data Muon's raw loss descends *faster* early but its task metric plateaus at holdout composite 0.846 (by step 3–4.5k, conversational schema collapsing) versus AdamW's 0.985. The cause is structural: rank-16 LoRA factors (1024×16) are too rectangular for the 5-step Newton-Schulz iteration to orthogonalize (empirically $\|O^\top O - I\|$ off-diagonals $\approx$0.5), and orthogonalizing the two factors separately is not the same operation as orthogonalizing their product. We keep AdamW.

E Reproducibility

Hardware: one RTX Pro 6000 (Blackwell), shared with an unrelated co-tenant workload throughout (all GPU jobs wrapped in retry/resume; the causal training run survived four OOM kills via checkpoint resume with $\leq$750 steps lost each). Training: LoRA r=16, α=32, batch 8, AdamW with the embedding/head weight-decay and cosine-schedule fixes of Appendix D; the unified checkpoint is selected at step 7.5k on the probe axes (dictation, context, intrusion) rather than the holdout composite alone. Evaluation and serving: vLLM 0.19, `skip_special_tokens=False` (the markers are reserved special tokens and are otherwise silently stripped — a gotcha worth one sentence). Code, the label-spec builder, the dictation/spelled probes, the replay benchmark, and the merged unified checkpoint are released at `https://github.com/19PINE-AI/turn-aware-asr`; the repository README documents the mapping from its internal checkpoint directory names to the recipes described in this paper.